

Supporting Anaesthetists during ‘Red Blanket’ Trauma Surgery: An Analysis of Work Practices and Requirements for a Head-Worn Display Support System

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ABSTRACT

‘Red blanket’ (RB) processes accelerate the surgical treatment of patients who have suffered severe trauma. Although the rapid start of surgery has positive effects on patient outcomes, the short time available to prepare for a case and the limited information available at the outset of care can be challenging for physicians. To support the lead of an RB team in making the right clinical decisions quickly, this work proposes a head-worn display (HWD) based information system. HWD could be valuable in RB settings, because they provide information in a continuous and hands-free manner, potentially allowing physicians to access case relevant data while treating the patient. Through an interview study with seven anaesthetists, we studied work practices of RB team leads and identified user requirements for the HWD based support system. We discuss opportunities and challenges for HWDs in RB settings and provide a detailed description of the developed HWD application.

CCS CONCEPTS

• **Applied computing** → Life and medical sciences; Health care information systems.

KEYWORDS

trauma, anaesthesia, red blanket, head-worn display

ACM Reference Format:

Paul Schlosser, Penelope Sanderson, Alvin A. Obed, and Christopher Futter. 2022. Supporting Anaesthetists during ‘Red Blanket’ Trauma Surgery: An Analysis of Work Practices and Requirements for a Head-Worn Display Support System. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI ’22 Extended Abstracts)*, April 29–May 05, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3491101.3519745>

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CHI ’22 Extended Abstracts, April 29–May 05, 2022, New Orleans, LA, USA

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ACM ISBN 978-1-4503-9156-6/22/04...\$15.00
<https://doi.org/10.1145/3491101.3519745>

1 INTRODUCTION

Car accidents, falls, gunshots, and similar incidents can cause injuries that are so severe that immediate surgery is required to stop bleeding and keep the patient alive [16, 17]. To allow for the rapid start of the necessary interventions, major hospitals have introduced procedures that enable the direct transfer of the patient from the helicopter landing site or ambulance bay into the operating theatre, minimising time spent in the emergency department (ED) where the patient usually undergoes initial assessment and treatment. In the Royal Brisbane and Women’s Hospital, where our study is situated, the expedited surgical treatment of severe trauma patients is called the ‘red blanket (RB) process’.¹

The present research explores which types of information anaesthetists require during RB surgery and how this information could be presented on a head-worn display (HWD). HWDs are wearables that can superimpose data over the user’s visual field [5]. Compared to conventional screens, HWDs might confer benefits in the context of RB surgery because they provide information in a continuous and hands-free manner. By enabling anaesthetists to access case relevant data while simultaneously treating the patient, HWDs might accelerate and improve care decisions.

In the remainder of the paper, we summarise related research and outline the novelty of our work. We then present the analysis of an interview study with seven senior consultant anaesthetists, and we detail properties of work during RB cases. Based on our results, we describe design tensions for the application of HWD in RB environments and outline the development of an interactive prototype. We conclude with an outlook on future research.

2 RELATED RESEARCH

After the arrival of ambulance staff at a trauma incident, a report of the patient’s status is usually transmitted to the closest major hospital and the patient is transported there. At the hospital, ED staff take over patient care with the goal of identifying and treating life-threatening injuries. This process is called trauma resuscitation [16]. As part of the resuscitation, staff carry out a primary survey during which they check the patient’s airway, breathing, blood circulation,

¹In the US the term ‘red blanket’ is also used to describe a hospital patient who receives special treatment due to their wealth or status as a celebrity. We do not discuss this phenomenon in the current paper.

and neurological functions. In a secondary survey, clinical tests are carried out and imaging data can be collected [1].

In the prehospital environment or during trauma resuscitation, it can become apparent that the patient requires immediate surgery to be physiologically stabilised, also known as damage control surgery. Because previous research has demonstrated that a rapid commencement of surgery has positive effects on trauma patient outcomes [6, 7], certain hospitals have implemented RB pathways that minimise or eliminate the time spent in the ED [2] for critically unwell patients. In the collaborating hospital, the RB pathway is activated if a hypotensive trauma patient is not responsive to fluid or other therapy. As a consequence of the RB activation, the patient bypasses the ED and resuscitation is performed in an operating theatre (OT) where surgery can commence.²

The RB pathway has not previously been studied from a human computer interaction (HCI) perspective. However, there is a broad corpus of CHI and CSCW studies that have explored the work practices of trauma teams in EDs (see e.g., [12, 18, 18, 24, 26]). Although compared to RB processes regular trauma resuscitation does not entail surgery and is slightly less time pressured, staff in both contexts probably use similar strategies to manage complexity. Particularly relevant for this paper is the research of Sarcevic and colleagues [20] who demonstrated that trauma teams in EDs rely extensively on verbal exchange to coordinate actions and memorise important details about a case. However, despite the skilled use of ‘transactive memory’ it can be challenging to maintain temporal and situation awareness during trauma resuscitation. Consequently, to support staff who attend trauma cases, Kusunoki and Sarcevic [15] developed peripheral information displays that can be accessed through large screens within the trauma bay. The displays present data such as patient demographics, findings, treatments, clinical test results, and stop clocks for different processes. In iterative evaluation sessions, physicians rated the system positively and highlighted the value of the integration of different information sources and time keeping tools. Fitzgerald and colleagues [10] developed a similar display and demonstrated in a clinical trial the system’s ability to reduce clinical errors and morbidity. However, in RB settings we believe that accessing information and support systems only via conventional wall mounted displays could be challenging because often “teams have their eyes and hands busy with the patient and instruments” [20].

HWDs might provide RB staff with relevant information effectively during surgery, as HWDs allow quick shifts of attention between patient and information source. The potential of HWDs in clinical settings has been explored in various studies [9, 25]. For example, HWDs have been used to guide manual tasks, enable remote collaboration, and support the monitoring of vital signs [22]. The application of HWDs for trauma teams has been previously suggested by Wu et al. [26]. However, to our knowledge this is the first study to develop a functional prototype of an HWD-based support systems for a trauma setting in general and an RB setting in particular.

With the present work we address two research questions: (1) What are work practices, challenges, and information requirements

of staff during RB cases, and (2) how could a HWD application support decision making by supporting the identified patterns of work? By answering these research questions, we hope to introduce the HCI community to the basic properties of work in RB environments and highlight the opportunities and risks of HWDs in that space. Furthermore, the design of the HWD application described herein lays the foundation for the evaluation of wearable devices in RB settings. The outcomes of future simulation-based studies and clinical trials that might be based on this research could have a positive impact on patient outcomes.

3 STUDYING WORK PRACTICES DURING RED BLANKET CASES

In the following, we address our first research question and describe the results of an interview study with senior consultant anaesthetists who regularly attend RB cases as team leads. We focused on team leads because previous studies have shown that they require access to the largest number of information sources [19], and therefore might benefit the most from a support system.

3.1 Method and Participants

We interviewed 7 senior anaesthetists who had previously led RB cases. All participants had more than 10 years of experience working as an anaesthetist. None of the participants had used a HWD before. Therefore, prior to the interviews we provided a short verbal description of HWDs to participants. The description lasted 5 minutes and participants were introduced to different HWD models available (e.g., optical see-through, video see-through), interaction mechanisms (e.g., voice, gestures, buttons), and common use cases (e.g., remote support, environment augmentation). During the interviews, we followed a semi structured approach. We asked participants predefined questions from an interview guide (see supplemental material) but deviated from it to ask for more details or clarification. All interviews were conducted by the first author, audio recorded, and lasted on average 42 minutes.

We used a reflective thematic analysis approach [3, 4] to generate insights from the collected data. First, we transcribed all interviews and assigned labels to participants’ statements (labels were developed inductively and without the use of a ‘code book’). Second, we grouped similarly labelled statements and looked for common patterns within our data. Third, we restructured our initial groupings and generated final themes that described and integrated the collected data. Finally, we analysed connections among themes and selected quotes for presentation in this manuscript. The data analysis was carried out primarily by the first author, but at several points throughout the process, authors CF and AO reviewed preliminary outcomes and provided alternative interpretations of data based on their clinical knowledge.

3.2 Results

Below we present three themes that we identified through our analysis (each theme is represented by one subsection). First, we focus on processes that allow for the rapid commencement of RB surgeries. Second, we describe challenges and work practices related to the ad-hoc structure of teams. Third, we describe challenges and work practices related to the management of time and information.

²For a detailed visualization of how regular trauma and RB pathways differ, see supplemental material.

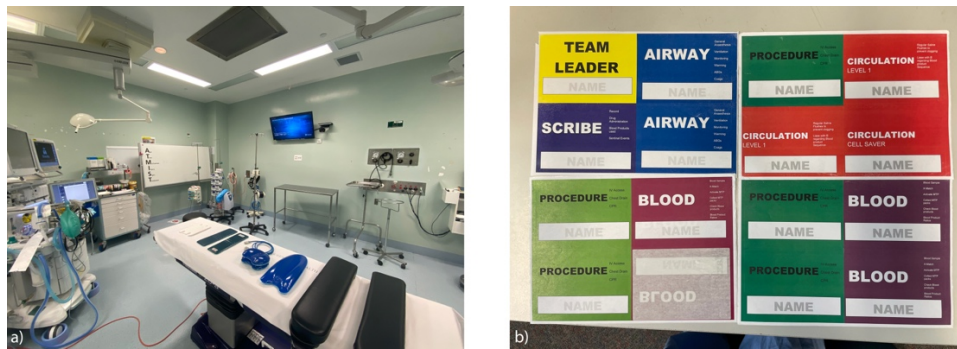


Figure 1: a) OT1 with patient data white board and the camera system; b) Labels that are used for role assignment

The latter two themes had a particular relevance for the design of the HWD application presented herein. Given the limited extent of the present format and our sole reliance on interview data, the following should be considered a focused analysis instead of an exhaustive ethnographic description of RB work.

3.2.1 Rapid commencement of RB surgeries. During the interviews, P1, P2, and P5 highlighted that RB cases occur relatively seldom—only 20–30 times a year. Therefore, it is impossible to have a team of physicians on standby who attend RB cases only. To facilitate the rapid commencement of surgery under these circumstances, staff rely on specific processes and structures within the hospital. P2 highlighted that a chain of informing had been implemented that enables the rapid alerting of multiple clinical disciplines should an RB case occur: “There is a specific number that goes to the general surgical consultant, and then to the coordinating anaesthetist, and theatre staff. [...] ED would make this one call and then all people would get notified.”

Once alerted, staff move quickly to OT 1 which is specifically set up for RB cases (see Figure 1a). P5 explained: “There [in OT 1] we have some setups that are specifically tailored to trauma surgery. So, they [the nursing staff] would be getting some laparotomy pack or potentially thoracotomy pack. Potentially, even opening up that equipment, depending on how close the case is”. The availability of two ultrasound machines in OT 1 (as opposed to one between multiple OT) was also described as an important feature to speed up the commencement of surgery upon arrival of the patient.

In addition to the RB alert system and the specific setup of OT1, including administrative staff in RB cases was described as crucial for the quick start of surgery. P1 outlined that each patient had to be assigned a unique identifier—the so-called unit record (UR) number. Without the UR number, lab-based tests, imaging, and other hospital services cannot be requested. Whereas during regular trauma cases, specific patient related data must be known to generate and assign the UR number, for RB cases administrative staff are instructed to generate a UR number for ‘Unknown Unknown’ and update the identifying data after the completion of surgery.

Finally, once the patient arrives in OT 1, an accelerated handover protocol is used by ED or ambulance staff to relay patient information to the RB team. P5 stated: “Sometimes this is not the most complete handover. Deliberately it’s meant to be very focused because time is of the essence. But obviously it is a balance, because

the handover is very important to get an understanding of what clinical information is available and what treatments have been started.” Acknowledging the importance of the handover, P3 explained that emergency department and ambulance staff were often asked to stay in the OT during the initial phase of the surgery to answer further questions and make sure that interventions carried out aligned with their understanding of the patient’s status.

Despite the above, all participants emphasized that the range and diversity of injuries that constitute RB cases make detailed preemptive planning impossible. Furthermore, participants outlined that the rapid commencement of RB surgery also introduces challenges. In the following section we will describe these challenges along with strategies that staff employ to manage them.

3.2.2 Ad-hoc structure of RB teams. All participants commented that a considerable challenge during RB cases was the establishment of a team that could handle the case, despite not having worked together in that specific team composition before. The anaesthetic coordinator who receives the call about the incoming RB case usually assigns the role of team lead to the most senior anaesthetist on duty. The team lead then gathers another 2–3 anaesthetists who will be responsible for different tasks during the RB surgery. Depending on the time of the day, the assembly of the RB team can vary in difficulty. P2 stated: “[Once you have been assigned the role of team lead] you have probably less than five minutes to get things organised in a true urgent fashion. The first priority is to call together a team. And usually, the team is a mix of people of some seniority. In the daytime you might be lucky, and you get another consultant or two. In the night-time, it would probably only be the senior registrar [being responsible for all anaesthetic duties initially] and the consultant being called in [from home]”.

The number of anaesthetic staff available affects the responsibilities of the team lead. Whereas during the day the team lead is supposed to step back, coordinate the case, and communicate with the different medical specialties in the room (P1, P2), during the night the team lead must get directly involved in the patient care (P6). Team leads are further responsible for controlling the number of staff in the OT, and the related noise level. P3 reported: “you have to make sure you don’t clutter the room with unnecessary people. Particularly because you’ve got so much equipment in there.” The workload of the team lead was described to be extremely high at all times.

To ensure that the team remains effective despite its ad-hoc structure, team leads employ a variety of formal and informal strategies. P1 described how a set of adhesive labels was used during RB cases to explicate and clarify responsibilities (see Figure 1b): *“You need a team structure. There’s allocation stickers [in OT 1]. So essentially people come in, they put a sticker on them [that tells everyone else their role], ‘you’re breathing’, ‘you’re drugs’, ‘you’re circulation’. And that’s their job.”* Furthermore, all participants expressed that during RB cases they constantly requested updates from other anaesthetists in the room, making sure that tasks that had been previously assigned were carried out or adjusted if the status of the patient had developed. P4 explained: *“If I am leading the team and I tell someone: ‘Can you give three units of blood’ what should be happening is the person coming back to me saying: ‘I have just given three units of blood’ So you close the loop! That’s effective communication in that crisis.”* P3 also described the deliberate interruption of all procedures within the OT at regular intervals to make sure that everyone was aware of the actions currently being carried out.

Based on the previously described work practices, in collaboration with the participants, we specified the following requirement for the development of a HWD application. To be functional in the RB environment, a support and information system should:

- communicate information mostly visually (instead of auditorily) due to the high noise level (REQ1);
- allow control with and without the use of hands, depending on involvement in interventions (REQ2);
- be usable by anaesthetists with different backgrounds and levels of expertise (REQ3);
- not increase workload and distract in particularly busy situations (REQ4).

3.2.3 Management of time and information. The information available about a patient and their injuries evolves throughout a RB surgery. Prior to surgery physicians might only know the age and sex of their patient as well as the injury mechanism (e.g., young male, single vehicle road traffic accident). To develop a better understanding of the patient’s current condition, the staff in OT 1 have access to a camera system that displays footage from the trauma bay in the ED (if the patient is transported to the OT directly from an ambulance/helicopter this system is not used). Although all participants who mentioned the system considered it as generally useful, extracting relevant information from the video feed was described as difficult, by P6: *“[On the video feed] we cannot see the vital sign monitor for their obs, and it is difficult to hear what’s happening [...] and it’s a bit tricky to work out what is going to be needed.”*

Once the patient arrives in OT1, three more clinically relevant information sources become available to the team lead and staff that can be used to inform clinical decisions. First, the information provided during the initial patient handover is recorded on a specifically setup whiteboard so that it does not get lost: *“We would have the whiteboard and we would write on it the facts about the case that we know. So, we would have a scribe. All they do is record what is happening. Because otherwise there is nothing recorded.”* (P1) Moreover, the vital sign monitor and the results of blood tests are important tools for decision making. The vital sign monitor is easily accessible when standing next to the patient but can be difficult to read when coordinating a case during the day from further back in

the room. Moreover, to see the results of clinical tests the team lead must move to a computer terminal that is positioned away from the patient. P2 reported: *“That [the results of the ROTEM test that the team requests in the OT] is always a pain to get, because you have to log onto the computer to get that”* Furthermore, participant P7 remarked that there was no system in place that reminded them to request tests or that notified them when results were available.

Generally, participants described time keeping as challenging but crucial for multiple processes. P3 described that there was a clock in OT1 but no timers *“So often, with these critical injuries, you get kind of absorbed in what is going on and you often don’t have a big overview of how long things have been going on for. So either a timer of the event since they [the patient] entered theatre or even a stop start kind of timer [would be useful to have], so that you can keep track of the duration of things.”* P1 also brought up the introduction of a timing system but highlighted the following: *“It [timer] shouldn’t be exclusively to the team leader. The information should be available to everyone. That way people can go ‘oh yeah look’. Because what can happen is it may prompt you: ‘The ROTEM [a clinical test] is due [to be requested]’, but you’re caught doing something. [...] Information isn’t exclusive, information is for everyone.”*

The information and time keeping needs of staff had the following implications for the development of the HWD application. Based on the above, to be functional in the RB environment, a support and information system should:

- integrate different information sources (REQ5);
- provide temporal awareness (REQ6);
- notify in case of patient developments and availability of test results (REQ7);
- remind staff when periodic actions are required (e.g., the request of tests) (REQ8);
- be accessible to all staff in the OT (REQ9).

3.3 Implications

Our analysis underlines that the work of RB team leads is high-tempo and highly information-dependent. The development of a system that simplifies information access and supports decision making seems valuable and justified based on the data collected. With regards to HWD as a hardware platform for such a system, we identified two design tensions—that is, properties of HWDs that could have both positive and negative effects in RB environments.

First, participants highlighted that they wanted to be prompted about developments but that their workload was already very high during RB cases. In the past, HWD have been demonstrated to be effective in attracting attention, especially if coupled with auditory stimuli [13]. However, they also had tendencies to distract [21]. It is unclear, at this point, whether the advantages of continuous informing can outweigh the disadvantages of potential distraction. Second, participants reported situations in which they needed to access information in a hands-free way (especially at night) but that it was also important to share information with everyone else in the OT. In previous research, the effects of HWDs on information sharing capabilities and team processes were mixed [8]. It is therefore unclear to what degree HWDs would isolate team leads in RB settings or allow them to share information with their colleagues.

To address the design tensions outlined, the combination of conventional screens mounted in the OT with a HWD that is worn by the team lead should be considered. This approach would allow the team lead to disable the HWD if distractions must be avoided, while still being able to check the conventional screens occasionally to remain aware of recent developments. Furthermore, a combination of HWD and conventional screens would support the dissemination of information in the OT. We are planning to develop such a 'combined system' in the next phase of the research project. We will conduct another series of interviews in which we study the information needs of surgeons, anaesthetists (taking on another role than team lead), and nurses. Based on the results of our analysis we will determine how HWDs and wall mounted screens could be integrated, and how information should be distributed across devices.

However, prior to pursuing the design of a combined system, we believe that it is valuable to complete the aim introduced at the beginning of this paper: Design a solely HWD based support and information system for the team lead of RB surgeries. The evaluation of a purely HWD based system will allow us to determine whether the anticipated mixed effects of the HWD materialise, or whether they can be mitigated through good user interface design [14], leading to quicker care decisions and the accelerated detection of deterioration events compared to the status quo. If a standalone HWD system could effectively support team leads during RB surgeries, this would also have important implications for field-based RB settings (e.g., military) where the setup of a combined system might be infeasible.

4 DEVELOPMENT OF A HWD BASED SUPPORT SYSTEM

In the following we address our second research question and describe how we translated the previously identified user requirements (REQ1-9) into a functional HWD prototype. The development process was divided into three broad phases. First, we generated multiple design ideas for HWD based support systems for RB team leads. The design ideas included systems that track tasks in a To-Do list fashion, present events in a timeline, provide basic notifications, or support case diagnosis in a similar fashion as described by Grundgeiger et al. [11]. We captured potential effects of the design ideas in storyboards and visions and collected feedback on them from 4 of the 7 individuals who had participated in the interview study. Second, we combined multiple elements of the design ideas and translated them into a single paper prototype (see Figure 2a). We collected feedback on the paper prototype from multiple members of the department of anaesthesia at the collaborating hospital during a project presentation. Third, in a final step, we developed a functional prototype of the system for the Vuzix M400 HWD and refined it through further input from clinicians. The Vuzix M400 is a monocular opaque HWD and is depicted in Figure 2c.

Having described the development process of the HWD application, in the following we outline its functionality. The application consists of a general view (see Figure 2b) and a detail view. In both views, a header is visible that shows the UR number of the patient, the current time, and time elapsed since the patient entered the OT, thereby providing basic temporal awareness (REQ6). The

general view combines multiple information sources (REQ5) that are integral to surgical settings and are likely to be relevant for anaesthetists independent of their experience level (REQ3).

The top half of the general view presents the vital signs of the patient as numerics and in waveforms. The bottom half features a chart that shows trendlines (development of vital signs over time), and a list with received test results (white) and open and completed tasks (yellow and grey backgrounds respectively; REQ8). The items in the list are related to the chart through horizontal lines, so to indicate when they occurred and to further boost temporal awareness (REQ6).

To notify the user about vital sign alarms, the receipt of clinical tests, or the presence of an uncompleted task, an auditory alarm is played through the earpiece of the HWD (the alarms can be muted and the entire HWD can be moved out of the visual field to avoid distraction; REQ4). A blinking dot on the HWD indicates which piece of information has been updated so to keep the application usable if sound cues are disabled or inaudible (REQ1). To interact with the HWD, users can use voice commands (REQ2). By saying "Hey assistant [activation word for the application], show test 1" the detail view for the specific clinical test would be activated. Similarly, through the phrase "Complete todo 1", participants could mark a task as completed. An alternative button-based input mechanism was implemented in case of voice recognition failure in the loud OT environment.

Currently, the HWD application is setup to receive the real-time vital signs from a clinical simulator in a similar fashion as described by Schlosser et al. [23]. To-do items and clinical test results are predefined and appear automatically at a specific time or per command of the experimenter in a 'Wizard of Oz' style. In an authentic clinical environment, tasks could appear based on a clinical algorithm. However, the current setup is optimised to facilitate the evaluation of the effects of the HWD in a future simulation study.

5 FUTURE WORK AND CONCLUSION

In summary, in the present paper we investigated the work practices and information requirements of team leads during the rapidly commenced surgical treatment of RB trauma patients and developed a HWD based support and information system for the environment. We described the mechanisms that allow physicians to treat their patients quickly after arrival at the hospital and highlighted challenges related to the ad-hoc assembly of teams and the evolving and distributed nature of information. Based on our findings, we conclude that key features of HWDs lend themselves well to the work practices in RB environments. However, the effects of potential HWD-induced distraction and wearer isolation are unclear. The HWD application that we designed in collaboration with the anaesthetists at the Royal Brisbane and Women's Hospital integrates vital sign information, clinical test results, and task reminders. It also focuses on the temporal awareness of staff by displaying events on a timeline.

In future research, we will evaluate the described HWD application and analyze its effects on clinical performance in a simulation study. We will also conduct further design activities to investigate ways of combining conventional screens with HWD for the purpose



Figure 2: a) Paper prototype; b) Functional HWD application; c) Participant providing feedback

of supporting trauma teams. Finally, we hope that the present paper encourages other researchers in the CHI and CSCW communities to engage more with the under-studied work environment of RB surgery.

ACKNOWLEDGMENTS

We would like to thank the anaesthetists at the Royal Brisbane and Women's Hospital for their participation in this study. We are grateful for the valuable feedback from Lisa Herbst on this manuscript. This work is supported by the Jamieson Trauma Institute and an Australian Commonwealth Research Training Scholarship.

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